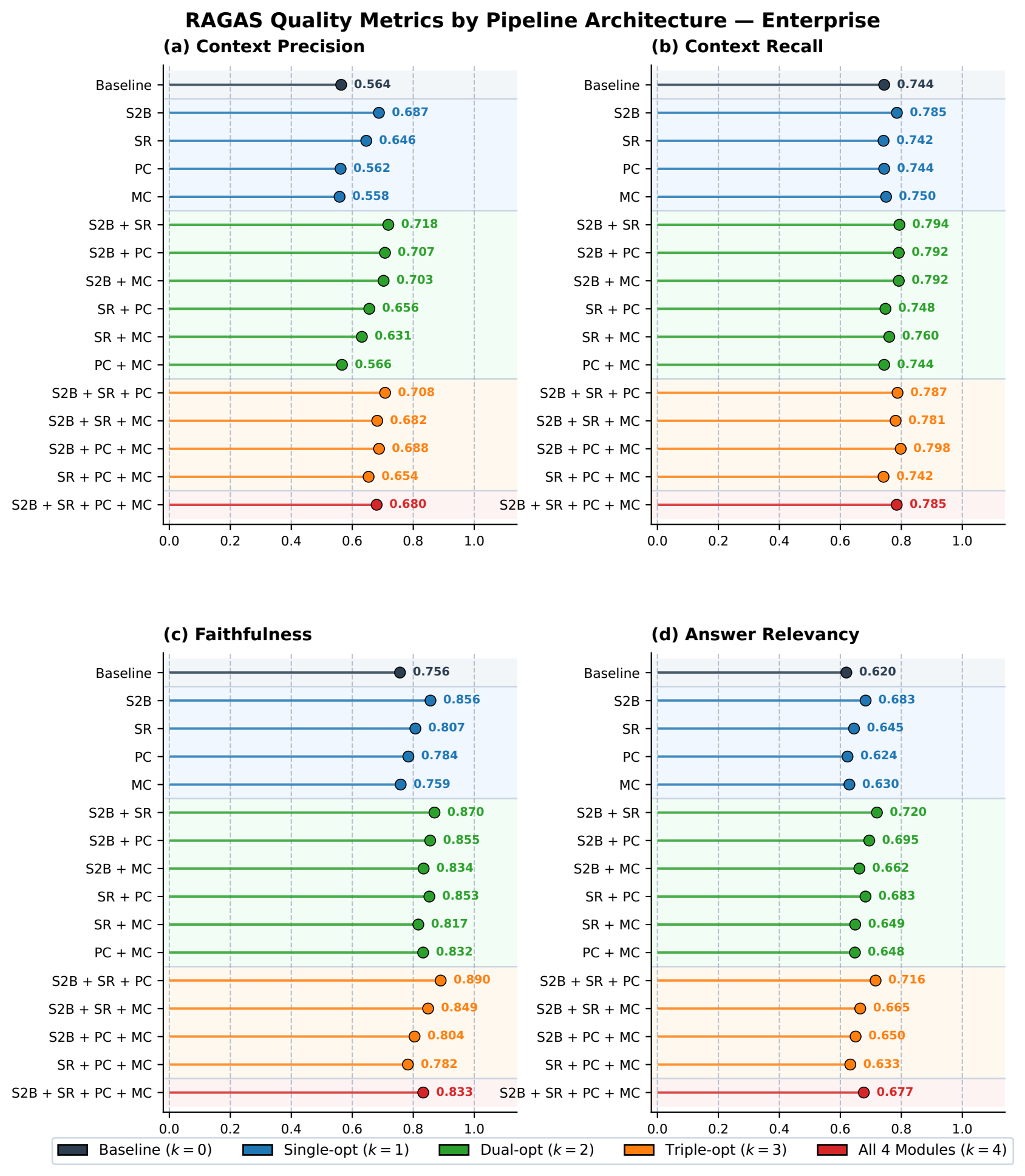
RAGAS Quality Metrics by Pipeline Architecture — Enterprise
(a) Context Precision
Baseline 0.564
S2B 0.687
SR 0.646
PC 0.562
MC 0.558
S2B + SR 0.718
S2B + PC 0.707
S2B + MC 0.703
SR + PC 0.656
SR + MC 0.631
PC + MC 0.566
S2B + SR + PC 0.708
S2B + SR + MC 0.682
S2B + PC + MC 0.688
SR + PC + MC 0.654
S2B + SR + PC + MC 0.680
0.0 0.2 0.4 0.6 0.8 1.0
(b) Context Recall
Baseline 0.744
S2B 0.785
SR 0.742
PC 0.744
MC 0.750
S2B + SR 0.794
S2B + PC 0.792
S2B + MC 0.792
SR + PC 0.748
SR + MC 0.760
PC + MC 0.744
S2B + SR + PC 0.787
S2B + SR + MC 0.781
S2B + PC + MC 0.798
SR + PC + MC 0.742
S2B + SR + PC + MC 0.785
0.0 0.2 0.4 0.6 0.8 1.0
(c) Faithfulness
Baseline 0.756
S2B 0.856
SR 0.807
PC 0.784
MC 0.759
S2B + SR 0.870
S2B + PC 0.855
S2B + MC 0.834
SR + PC 0.853
SR + MC 0.817
PC + MC 0.832
S2B + SR + PC 0.890
S2B + SR + MC 0.849
S2B + PC + MC 0.804
SR + PC + MC 0.782
S2B + SR + PC + MC 0.833
0.0 0.2 0.4 0.6 0.8 1.0
(d) Answer Relevancy
Baseline 0.620
S2B 0.683
SR 0.645
PC 0.624
MC 0.630
S2B + SR 0.720
S2B + PC 0.695
S2B + MC 0.662
SR + PC 0.683
SR + MC 0.649
PC + MC 0.648
S2B + SR + PC 0.716
S2B + SR + MC 0.665
S2B + PC + MC 0.650
SR + PC + MC 0.633
S2B + SR + PC + MC 0.677
0.0 0.2 0.4 0.6 0.8 1.0
Baseline ($k = 0$)
Single-opt ($k = 1$)
Dual-opt ($k = 2$)
Triple-opt ($k = 3$)
All 4 Modules ($k = 4$)